

actuators should be controlled. These actuators are placed in the steering wheel, gas pedal and brake pedal, and receive commands from the IR sensors," explains Natarajan M.M., vice president for South Asia and Bhartendu Mishra, director - marketing of Arrow Asia-Pacific in an interview with EFW.

"There are also IR cameras that also monitor the driver's subtle head movements. A central processor in the car translates the sensor input into motion commands for the car. It is aided by an onboard global positioning system (GPS) that updates 100 times per second, creating virtual boundaries, and provides data for the car's self-correction in certain cases," adds Natarajan.

An Internet of Vehicles

The Internet of Vehicles (IoV) seems to be a new concept based on the IoT. However, Huawei's website says that, nascent forms of the IoV are already in existence.

"Intelligent transport systems (ITS) in Europe and Japan have adopted certain forms of IoV technology. In New Delhi, all 55,000 licensed rickshaws have been fitted with GPS devices so that drivers can be held accountable for their questionable route selection. China's Ministry of Transport (MOT) had ordered that GPS systems be installed and connected on all long-haul buses and hazmat vehicles by the end of 2011 to ensure good driving habits and reduce the risk of accidents and traffic jams. The Brazilian government has set a goal for all cars in circulation to be fitted with electronic identification (ID) chips from its National Automated Vehicle Identification System (Siniav)."

Huawei also claims that the launch of the US National Strategy for Trusted Identities in Cyberspace (NSTIC) is a milestone for IoV, as it requires that security chips be embedded in all online devices, including those in vehicles.

The telecom connection. In India, Vodafone has partnered with MREV

to provide machine-to-machine (M2M) communication services for the e2o range of cars and a central application. A *Times of India* report mentions that, AT&T claimed about 20 million connected devices from cars to cargo ship container sensors in 2014, up 21 per cent from the year earlier. While it has not yet revealed its revenue from its IoT business, the fourth quarter of 2014 saw AT&T adding 800,000 connected cars out of 1.3 million connected devices in its network.

Some exciting technologies

The Mercedes-Benz F105 car we mentioned in the beginning has another impressive feature, in that, its windows are almost invisible from the outside while, the same panels function as TV screens from the inside.

Structural electronics certainly would not make the windows disappear, but it would allow integration of sensors and other components within the vehicle's body and undercarriage. The present implementation of this technology, although at a very basic level, is the highly-sensitive microphones in the very front and back part of some cars that record the structural sound, and in case of an accident, send the signal to the airbag controller.

Could structural electronics be the next big thing in cars, especially when it is biomimetic or imitative of nature?

"It is a fascinating and largely under-explored area right now. Commercial adaptation will probably start with the aeronautical sector—smartskins, printed electronics, structural super capacitors where the chassis can also act as both load-bearing and energy-storage structure," explains Sandeep Bairampalli, expert on robotics, Robert Bosch Engineering and Business Solutions.

Another innovative use of electronics is with charging. Qualcomm's recently announced Halo Wireless Electric Vehicle Charging (WEVC) technology provides wireless-charging

capabilities in a small vehicle package. This will allow users to charge their electric vehicles (EVs) easily and quickly. In addition to that, a supportive tech for Halo, named WiPower, enables consumer electronics to charge wirelessly in-vehicle.

We have all heard of the supercapacitor vs battery debate. Could replacing batteries with supercapacitors be feasible in cars?

"Theoretically, yes. All major organisations are working towards the realisation of this ideal scenario. One focus area of Bosch is electrification; how do we unlock the potential of energy for the benefit of life? Gradually, we will see this happening as the energy density of supercaps increases. We do not have to match the li-ion energy density to see significant market share being taken by supercaps with respect to li-ion, as the power density is much higher and these have significantly higher operational lives," adds Bairampalli.

Is all this safe? "Apart from hacking by third parties, there is a much bigger issue that is now stalling the overall smartcar development; it is cross border data transfer (CBDT).

By law, customer data cannot reside outside the country. And this is a somewhat bigger impediment. Most advanced car manufacturers such as VW, Audi, Mercedes and Toyota do not have local IT infrastructure, and it is usually based at their headquarters (outside India). This means, in order to just make the basic system fully-compliant with law, there is a longer process of infrastructure set up. Then, sharing customer data or even customer's car data with any other customer and privacy concerns raised from those sharing are next-in-line issues," explains Anand.

Is KITT here

Who knows, years down the line we might have something like KITT from the 1982 television series Knight Rider, a robotic car with enough artificial intelligence (AI) that would help it pass the Turing test! ●